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Life cycle assessment of electrolytic manganese metal production

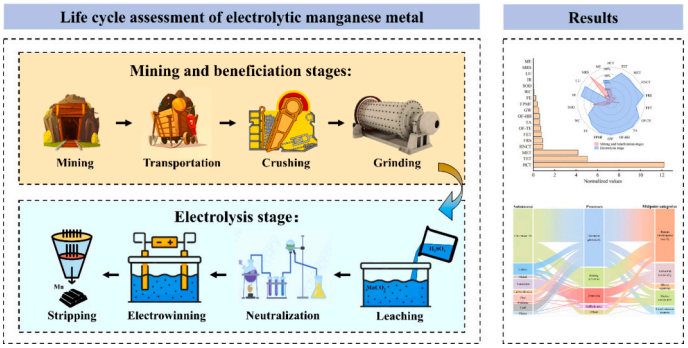
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HIGHLIGHTS

- LCA was conducted on the production of electrolytic manganese metal.
- Human carcinogenic toxicity is the most prominent impact category.
- The key process of environmental impact is electricity generation, accounting for 57.43 %.
- Chromium (VI) contributes to Human carcinogenic toxicity by as much as 91 %.

GRAPHICAL ABSTRACT



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ABSTRACT

Manganese is an indispensable metal widely used in various fields. China ranks as the fourth-largest producer of manganese ore and the largest producer of electrolytic manganese metal (EMM). However, EMM production is linked to high energy consumption and pollution. This study conducts a life cycle assessment (LCA) of EMM production in the Manganese Triangle region of China to comprehensively evaluate its environmental impact. Results show that Human carcinogenic toxicity, mainly from electricity generation (65.3 %) and mining activities (24.4 %), is the most significant environmental impact. Chromium (VI) is identified as the predominant hazardous substance, contributing up to 91 % to Human carcinogenic toxicity. Endpoint results estimate that the production of 1 t of EMM results in 1.01E-02 DALY of harm to human health, 1.97E-05 species.yr of harm to the ecosystem, and $227.15 worth of resource depletion. Simulation scenarios demonstrate that replacing thermal power with hydropower can reduce environmental pollution by over 90 %. Finally, based on the findings, technical measures for promoting clean production of EMM were proposed.

1. Introduction

Manganese, a versatile metal, finds applications across various industries such as metallurgy (Sun et al., 2022), battery manufacturing (Song et al., 2023), and biomedical fields (Sisakhtnezhad et al., 2023). Currently, the increasing global demand for electric vehicles is driving the demand for power batteries, thereby boosting the demand for manganese metal (Maisel et al., 2023). The 2022 data on China's

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manganese ore consumption suggests a notable increase in the utilization of manganese in lithium-ion batteries, surpassing the 1 % mark (IMNI, 2023). China not only ranks as the world's fourth-largest producer of manganese ore, following South Africa, Gabon (USGS, 2023), and Australia but also holds the top position as the largest producer of electrolytic manganese metal (EMM) (D.J. He et al., 2021; Li et al., 2023). The current situation is that large-scale EMM production enterprises only exist in China and South Africa. Since 2008, China has consistently contributed to over 98 % of the global production of EMM, establishing itself as the leading producer worldwide (Han and Wu, 2019).

The electrolysis production of manganese metal represents a processing industry characterized by elevated resource and energy consumption, substantial pollution generation, high costs, and relatively low profits (Ning et al., 2010). The majority of EMM production enterprises in China are concentrated in the renowned Manganese Triangle, delineated by Hunan, Guizhou, and Chongqing. However, enterprises in the region lacked environmental awareness, resulting in significant environmental damage and pollution caused by these enterprises (S.C. He et al., 2021). The pollution generated by industrial production of EMM, including electrolytic manganese residue, the release of Mn^{2+}, and ammonia nitrogen, poses a serious threat to ecosystems and human health (Chen et al., 2022). Thus, it is necessary to comprehensively assess the environmental impact of EMM production in the Manganese Triangle region in order to make recommendations for cleaner production.

Life cycle assessment (LCA) stands as a potent tool for environmental analysis, systematically quantifying the environmental impact of products, processes, or activities (ISO 14040, 2006). Widely applied across diverse sectors, including agriculture (van der Werf et al., 2020), construction (Rodrigues et al., 2023), and mining (Tao et al., 2022), LCA provides a comprehensive understanding of environmental implications. For the manganese metal industry, Westfall et al. (2016) collected primary data from 16 manganese alloy producers worldwide and conducted a LCA. The results of the study show that total GWP, AP, and POCP for 1 kg of average manganese alloy was 6 kg CO_2eq, 45 g SO_2eq, and 3 g C_2H_4eq, respectively. Ning et al. (2010) conducted a comprehensive material balance and pollutant analysis of the EMM industry in China, revealing the sources and destinations of major pollutants. However, their study did not address the aspect of climate change. Peng et al. (2011) utilized LCA to compare the environmental impacts of EMM production in China and South Africa, focusing solely on the global warming potential and acidification potential. Davourie et al. (2017) conducted a LCA to evaluate particulate matter emissions in manganese alloy production. The results revealed that 66 % of production-related particulate matter emissions occur outside of manganese facilities, with direct or on-site emissions accounting for 34 % of the total PM emissions. However, it only studied particulate matter and did not investigate other pollutants. Farjana et al. (2019) conducted an LCA analysis on manganese ore production based on the Ecoinvent database, but did not cover the subsequent electrolysis part of manganese concentrate. Zhang et al. (2020) conducted a detailed LCA analysis on EMM production in China, but the selected mining and electrolysis data were not sourced from the same enterprise, resulting in spatial discrepancies. Furthermore, the majority of existing studies pertaining to EMM have predominantly focused on the analysis of its pollutants. For example, Zhang et al. (2023) investigated the heavy metal pollution situation of Mn, Zn, Pb, and others in a northern Chinese EMM industrial area and its surrounding environment. Sun et al. (2020) utilized sulfur resource recovery from roasting electrolytic manganese residue and desulfurizing manganese ore for the clean production of electrolytic manganese. Xu et al. (2014) proposed a comprehensive wastewater treatment method for EMM production through a study on the water balance of EMM enterprises. In conclusion, China currently lacks a high-quality, comprehensive LCA study on the environmental impacts of EMM production. To address this, this study selected the most representative electrolytic manganese production company in the Manganese Triangle, the region with the richest manganese deposits in China. A LCA was conducted on electrolytic manganese production from the mining stage to the electrolysis stage.

This study aims to (1) analyze the environmental burden of EMM production in the Manganese Triangle region of China using the LCA method; (2) obtain quantitative results of life cycle impact assessment (LCIA) along with uncertainty information and identify the major impact categories on the environment; (3) identify key processes and substances, conduct sensitivity analysis on key processes, and examine the degree of variation in LCIA results; (4) simulate various energy scenarios, compare their corresponding environmental impacts, and propose optimization strategies for the electrolytic manganese metal production industry in China. After conducting a quantitative LCA, analyze from multiple perspectives, including contribution, uncertainty, and sensitivity, to comprehensively assess the current status and environmental impacts of EMM production technology in China. This analysis aims to optimize resource and energy consumption and reduce pollutant emission intensity. Propose technical measures to achieve clean and efficient EMM production and to build a green and sustainable energy system.

2. Scope and methods

2.1. Case study

A typical EMM production enterprise in the Manganese Triangle region of China, was selected for the study. The ore from this enterprise contains approximately 18.77 % manganese, with minimal industrial utility for other components. The mining method employed by this enterprise is the room and pillar mining method, while the production of EMM adopts the hydrometallurgical method. The production process is categorized into mining and beneficiation stages and electrolysis stage. Manganese ore is extracted from underground mines through digging and blasting, then lifted to the surface and conveyed to an on-site beneficiation plant for crushing. The finely crushed manganese concentrate is subsequently transported about 30 km to the electrolysis plant for electrolysis, resulting in a final product with 99.8 % EMM.

According to the enterprise's statistics, 1 t of raw ore is approximately transformed into 0.6 t of concentrate after processing, with 0.35 t used for tailings backfilling and 0.05 t lost. Among the 0.6 t of concentrate, 0.064 t are converted into EMM, 0.455 t become tailings, and 0.08 t are lost in various processes. This enterprise, situated in Guizhou Province within the Manganese Triangle region, plays a crucial role in China's manganese ore and EMM production. In 2022, Guizhou Province held the second position in China for manganese ore production and the third position for EMM production. The enterprise is a representative enterprise of Guizhou Province consequently, the study of this enterprise serves as a representative case for China's EMM production.

2.2. Functional unit and system boundary

For this study, the functional unit was set as 1 t of EMM, serving as the basis for all considerations, including raw materials, energy inputs, transportation, and emissions to air and water.

The system boundary for EMM production is depicted in Fig. 1. Utilizing the "cradle-to-gate" approach to define the system boundary, the entire system is segmented into two major parts: the mining and beneficiation stages, and the electrolysis stage. The mining and beneficiation stages encompass processes such as tunneling, support, hoisting, ventilation, crushing, and grinding. The electrolysis stage comprises processes like raw material leaching, neutralization, electrolysis, passivation, and rinsing. Tailings from the mining and beneficiation stages are treated in tailings dam, and wastewater is either recycled or discharged after treatment in wastewater sedimentation tanks.

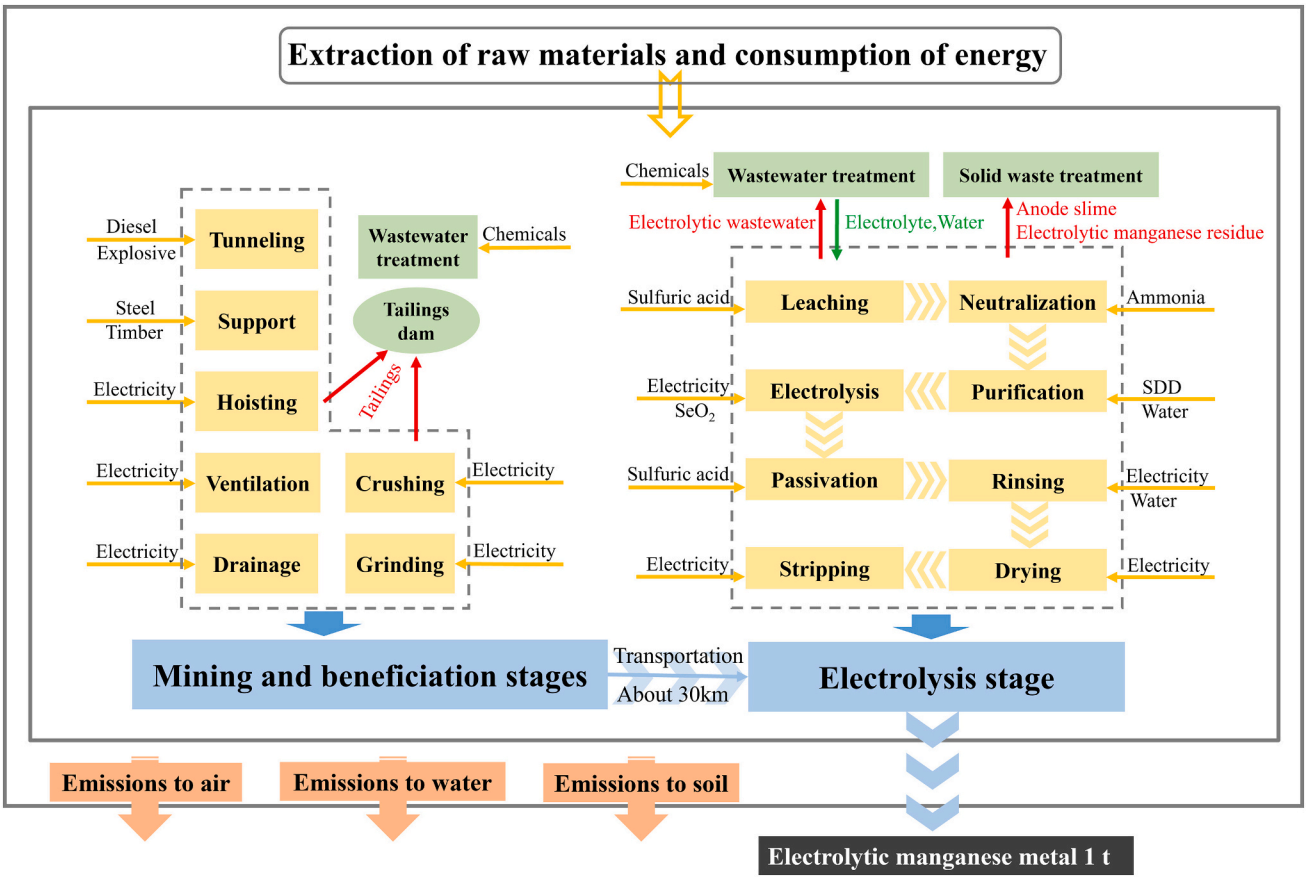


Fig. 1. System boundary of the EMM production life cycle.

Wastewater from the electrolysis stage undergoes treatment in a wastewater treatment plant before being reused in production. Industrial solid wastes, including electrolytic manganese residue, are recycled and reused following treatment. Domestic wastes are transported to an off-site landfill. Additionally, the system incorporates a service area that encompasses the daily activities of personnel. It is crucial to emphasize that processes such as equipment production and equipment maintenance fall outside the purview of this system boundary.

2.3. Life cycle inventory analysis

The main field data for this study were obtained from EMM producers in Songtao, Guizhou Province, China. The data for the mining and beneficiation stages comes from the company's manganese mine, while the data for the electrolysis stage comes from the company's EMM plant, located 30 km away from the manganese mine. In addition, some of the missing data from the site were obtained from the Ecoinvent database, recognized as one of the most complete and transparent life cycle inventory databases globally. Being one of the world's most comprehensive and transparent life cycle inventory databases, the Ecoinvent database supports the environmental assessment of products and processes worldwide (Ecoinvent centre, 2015).

To comprehensively evaluate the data, ensuring accuracy and reliability, we assume that the measurement data in LCA follow a lognormal distribution. A notable characteristic of the lognormal distribution is that the square of the geometric standard deviation (GSD^2) covers a 95 % confidence interval, and the 95 % confidence interval range can be obtained by multiplying and dividing the measured values by GSD^2. This can be expressed as

$$Probability\left\{\frac{m}{(GSD^2)} < X < GSD^2 \times m\right\} = 0.95 \quad (1)$$

where m is the measured data.

GSD^2 can be calculated according to the following formula

$$GSD^2 = exp\sqrt{[ln(U_1)]^2 + [ln(U_2)]^2 + [ln(U_3)]^2 + [ln(U_4)]^2 + [ln(U_5)]^2 + [ln(U_b)]^2} \quad (2)$$

where $U_1 \sim U_5$ are uncertainty factors for reliability, completeness, temporal relevance, geographic relevance, and technological relevance, and U_b is the basic uncertainty factor.$U_1 \sim U_5$ and U_b are determined by the Ecoinvent 3 pedigree matrix(Frischknecht et al., 2005).

The final life cycle inventory is shown in Table 1.

3. Results

3.1. Midpoint results

The LCIA results at midpoint levels were analyzed using the ReCiPe 2016 method, a widely adopted approach in LCA analysis (Huijbregts

Table 1
Life cycle inventory based on functional unit of 1 t of EMM production.

	Substance	Unit	Amount	GSD^2
Elements in ore	Manganese (Mn)	kg	2815.50	1.16
	Iron (Fe)	kg	51.45	1.31
	Phosphorus (P)	kg	29.85	1.31
	Copper (Cu)	g	936.00	1.32
Energy and raw materials consumption	Electricity	kWh	7675.32	1.16
	Diesel	kg	8.09	1.16
	Detonator	g	280.00	1.16
	Explosive	kg	10.88	1.16
	Steel	kg	36.00	1.24
	Rubber	kg	0.48	1.22
	Wood	kg	4.13	1.24
	Fresh water	t	6.22	1.22
	Tap water	t	2.46	1.22
	PAC	g	285.15	1.22
	Sulfuric acid	kg	982.35	1.33
	Ammonia	kg	72.15	1.24
	Sodium dimethyl dithiocarbamate (SDD)	g	76.64	1.57
	Selenium dioxide (SeO_2)	kg	1.31	1.24
Emissions to air	Ammonia	g	343.41	1.64
	Sulfuric acid	g	159.50	1.24
	Carbon monoxide (CO)	g	49.08	5.88
	Carbon dioxide (CO_2)	kg	1.21	1.63
	Particulate matter (PM10)	g	3497.96	1.52
	Nitrogen oxides (NO_x)	g	258.56	1.88
	Hydrocarbon	g	73.24	2.11
	Sulfur dioxide (SO_2)	g	41.43	1.63
Emissions to water	Suspended solids (SS)	g	585.22	1.54
	Chemical Oxygen Demand (COD)	g	134.88	1.54
	Ammonia nitrogen (NH_3—N)	mg	46.21	1.54
	Iron (Fe)	g	3.19	5.12
	Manganese (Mn)	g	9.53	5.12
	Sulfide	g	3.53	1.63
Waste disposal	Solid waste sanitary landfill	kg	6.84	2.34
	wastewater treatment	t	7.81	2.34
	Tailings	t	9.24	1.25
Transportation	Lorry	t·km	154.60	2.06

et al., 2017). Table 2 displays the LCIA midpoint results across 18 impact categories. Take global warming, for instance; the production of 1 t of EMM results in an impact of 5214.56 kg CO_2 eq. It's crucial to note that this figure doesn't solely represent carbon dioxide; it encompasses various greenhouse gases like methane (CH_4), nitrous oxide (N_2O), and hydrofluorocarbons (HFCs). The choice of carbon dioxide as a reference in the global warming impact category is due to its typicality. Other greenhouse gases are converted to carbon dioxide equivalents using characterization factors. For instance, a factor of 1 for carbon dioxide and 25 for methane implies that each tonne of emitted methane is equivalent to 25 t of carbon dioxide in terms of its impact on climate change. Moreover, within the mining industry, there is significant concern about mineral resource depletion. LCIA midpoint results indicate that, concerning mineral resource scarcity, the production of 1 t of EMM would lead to the depletion of mineral resources equivalent to 268.97 kg of copper. Similar considerations apply to other impact categories.

Using Monte Carlo simulation for uncertainty analysis helps assess the uncertainty of various data and parameters in LCA, providing estimates of the reliability range of assessment results and offering decision-makers more comprehensive decision-making information. Therefore, uncertainty analysis on the midpoint results was conducted using the Monte Carlo method. This approach further enhances the robustness of the assessment by addressing uncertainties in both data and results. For instance, in the case of global warming, where the calculated environmental impact value is 5214.56 kg CO_2 eq and its GSD^2 value is 1.30, the 95 % confidence interval for global warming is determined to be 4011.20 kg CO_2 eq to 6778.93 kg CO_2 eq. Similar calculations apply to other impact categories, and the results can be found in Table 2.

Table 2
LCIA midpoint results (Based on each functional unit).

Category	Unit	Value	GSD^2	Value range
Global warming	kg CO_2 eq	5214.56	1.30	4011.20–6778.93
Stratospheric ozone depletion	kg CFC-11 eq	1.85×10^{-3}	1.34	1.38×10^{-3}-2.48×10^{-3}
Ionizing radiation	kBq Co-60 eq	5.94	4.04	1.47–24.00
Ozone formation, Human health	kg NO_x eq	14.90	1.15	12.96–17.14
Fine particulate matter formation	kg $PM_{2.5}$ eq	11.12	1.24	8.97–13.79
Ozone formation, Terrestrial ecosystems	kg NO_x eq	14.96	1.15	13.01–17.20
Terrestrial acidification	kg SO_2 eq	28.96	1.19	24.34–34.46
Freshwater eutrophication	kg P eq	0.18	2.66	0.07–0.48
Marine eutrophication	kg N eq	9.27×10^{-3}	1.26	7.36×10^{-3}-1.17×10^{-2}
Terrestrial ecotoxicity	kg 1,4-DCB eq	5725.81	1.68	3408.22–9619.36
Freshwater ecotoxicity	kg 1,4-DCB eq	1.06	2.50	0.42–2.65
Marine ecotoxicity	kg 1,4-DCB eq	4.73	2.47	1.91–11.68
Human carcinogenic toxicity	kg 1,4-DCB eq	38.29	3.02	12.68–115.64
Human non-carcinogenic toxicity	kg 1,4-DCB eq	154.08	3.32	46.41–511.55
Land use	m^2a crop eq	67.93	1.70	39.96–115.48
Mineral resource scarcity	kg Cu eq	268.97	1.20	224.14–322.76
Fossil resource scarcity	kg oil eq	1048.96	1.19	881.48–1248.26
Water consumption	m^3	23.76	1.68	14.14–39.92

3.2. Normalized midpoint results

In LCIA, assessment results are typically represented by midpoint indicators of various environmental impact categories (such as global warming, acidification, toxicity, etc.). These indicators have different units and magnitudes, making them difficult to compare and interpret. To address this issue and identify the categories with the most detrimental environmental impact, normalization of these midpoint indicators is necessary (Zhou et al., 2022). This process ensures that the indicators are comparable by aligning them to a common scale. Fig. 2. illustrates the normalization of these results and provides a comparison between the mining and beneficiation stages and the electrolysis stage. As depicted in Fig. 2., human carcinogenic toxicity imposes the most substantial environmental burden among the 18 impact categories, with terrestrial ecotoxicity and marine ecotoxicity also exhibiting noteworthy impacts. Other categories demonstrate a more limited impact, characterized by relatively smaller standard values for marine eutrophication, mineral resource scarcity, land use, ionizing radiation, and stratospheric ozone depletion. Furthermore, upon comparing the mining and beneficiation stages with the electrolysis stage, it becomes evident that the electrolysis stage generally imposes a higher environmental impact. Specifically, in the human carcinogenic toxicity category, the electrolysis stage contributes to over 60 % of the impact, while in terrestrial ecotoxicity and marine ecotoxicity categories, the electrolysis stage accounts for over 80 % of the impact. Conversely, more than 90 % of the impacts related to mineral resource scarcity originate from the mining and beneficiation stages.

3.3. Grouped characterization midpoint results

To yield practical research outcomes, propose viable improvement measures for reducing environmental burdens, and guide the eco-

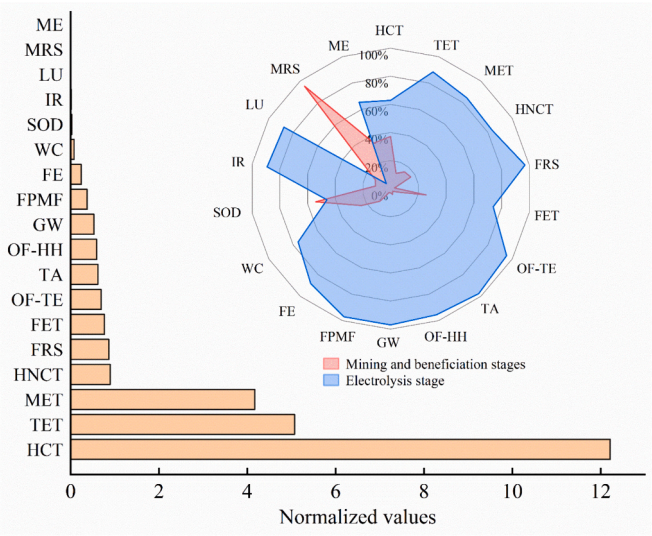


Fig. 2. Normalized midpoint results (HCT: human carcinogenic toxicity; TET: terrestrial ecotoxicity; MET: marine ecotoxicity; HNCT: human non-carcinogenic toxicity; FRS: fossil resource scarcity; FET: freshwater ecotoxicity; OF-TE: ozone formation-terrestrial ecosystems; TA: terrestrial acidification; OF-HH: ozone formation-human health; GW: global warming; FPMF: fine particulate matter formation; FE: freshwater eutrophication; WC: water consumption; SOD: stratospheric ozone depletion; IR: ionizing radiation; LU: land use; MRS: mineral resource scarcity; ME: marine eutrophication).

friendly production of EMM, the process of EMM production within the system boundary was further categorized into five groups: mining activities, electricity generation, ammonia, sulfuric acid, and others. Fig. 3. illustrates the distribution of these five groups across the 18 midpoint impact categories. Human carcinogenic toxicity is primarily attributed to electricity generation and mining activities, with electricity generation contributing 65.3 % to this category and mining activities contributing 24.4 %. Electricity generation predominates in impact categories such as global warming, ozone formation, marine ecotoxicity, and fossil resource scarcity. Mining activities account for over 90 % of the impact related to mineral resource scarcity. Sulfuric acid contributes more than 30 % to terrestrial acidification, freshwater ecotoxicity, marine ecotoxicity, and human carcinogenic toxicity, while ammonia plays a significant role in terrestrial ecotoxicity. Overall, electricity generation predominates in most categories, and this is a crucial aspect that requires particular attention.

3.4. Identification of key factors

In the normalized midpoint results, the top three impact categories, namely human carcinogenic toxicity, terrestrial ecotoxicity, and marine ecotoxicity, were identified as having the most significant environmental impacts, in that order. To further uncover the factors contributing to these key impact categories, a key factor analysis was conducted. Notably, in the key factor identification, an analysis was conducted on two additional impact categories: fossil resource scarcity and global warming. Fossil resource scarcity holds the fifth position among the 18 impact categories, considering its research significance (Tahir and Chen, 2021). Furthermore, global warming, being a subject of current global debate, was taken into account (Huang et al., 2017). The results depicting the identified key factors for these five categories are presented in Fig. 4. The key processes contributing to environmental impacts are electricity generation, mining activities, and ammonia consumption, accounting for 57.43 %, 18.91 %, and 14.92 % respectively. Chromium (VI) stands out as the most critical hazardous substance within the realms of electricity generation and mining activities, contributing significantly to human carcinogenic toxicity, with a substantial 91 % impact. In the context of electricity generation, copper plays a pivotal role, contributing 35 % to terrestrial ecotoxicity. On the other hand, coal emerges as a key player, accounting for over 99 % of the impact on fossil resource scarcity. The impact of global warming is predominantly driven by carbon dioxide, responsible for 82 %, with a smaller fraction attributed to methane and other substances. Notably,

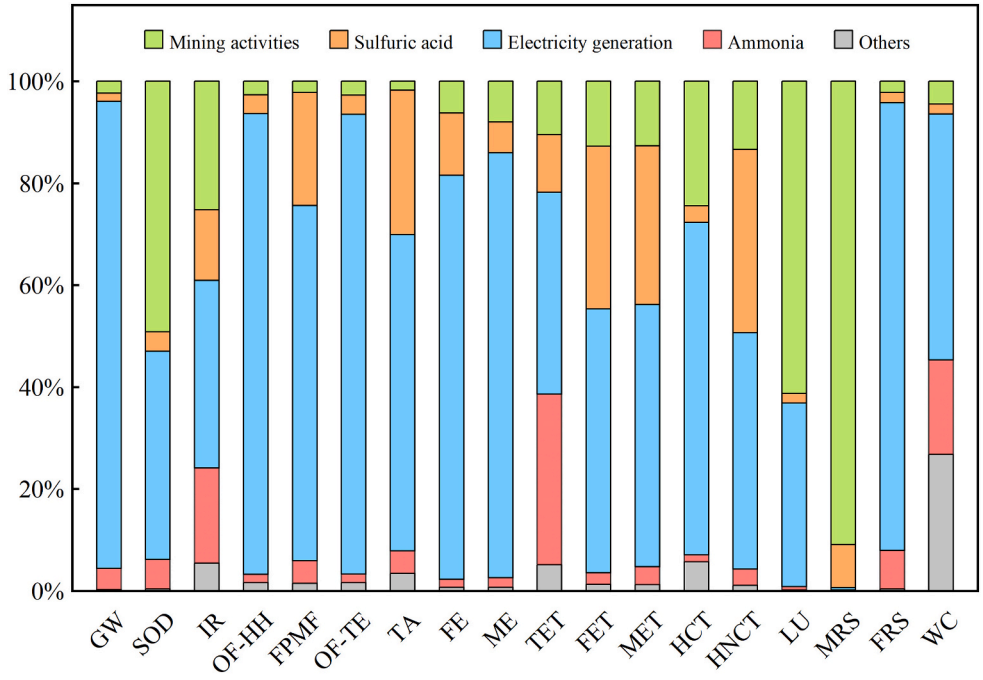


Fig. 3. Contributions of groups to midpoint categories (GW: global warming; SOD: stratospheric ozone depletion; IR: ionizing radiation; OF-HH: ozone formation-human health; FPMF: fine particulate matter formation; OF-TE: ozone formation-terrestrial ecosystems; TA: terrestrial acidification; FE: freshwater eutrophication; ME: marine eutrophication; TET: terrestrial ecotoxicity; FET: freshwater ecotoxicity; MET: marine ecotoxicity; HCT: human carcinogenic toxicity; HNCT: human non-carcinogenic toxicity; LU: land use; MRS: mineral resource scarcity; FRS: fossil resource scarcity; WC: water consumption).

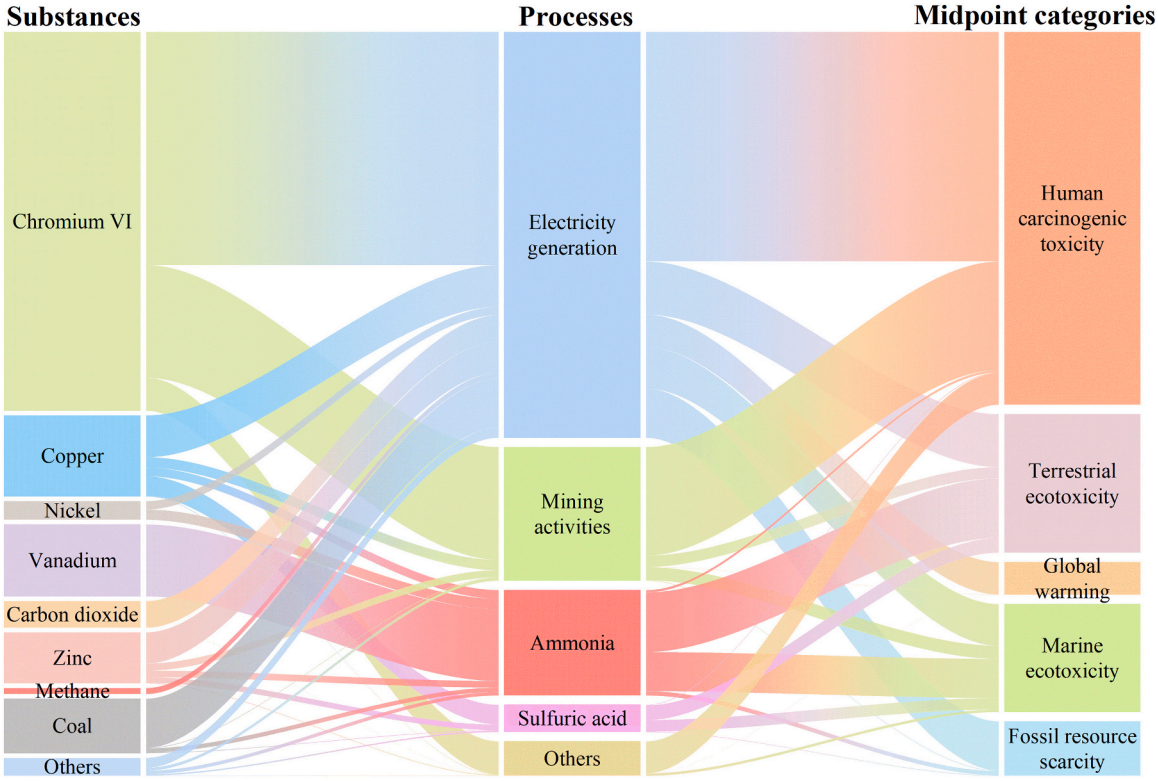


Fig. 4. Key factors analysis.

Table 3
LCIA endpoint results (based on each functional unit).

Category	Unit	Value	GSD^2	Value range
Human health	DALY	1.01×10^{-2}	1.24	8.15×10^{-3}-1.25×10^{-2}
Ecosystems	species.yr	1.97×10^{-5}	1.34	1.47×10^{-5}-2.64×10^{-5}
Resources	$	227.15	1.19	190.88–270.31

vanadium takes the spotlight as the primary pollutant in the production of liquid ammonia.

3.5. LCIA endpoint results

The results of the LCIA endpoints, as depicted in Table 3, classify environmental concerns into three endpoint damage categories: human health, ecosystems, and resources. These categories are expressed in terms of disability-adjusted life-years (DALYs), loss of species per year (species.yr), and monetary value (dollars), respectively. The purpose of this classification is to elucidate the magnitude of damage to each environmental theme. Table 3 illustrates that the production process of 1 t of EMM results in 1.01×10^{-2} DALYs of damage to human health, 1.97×10^{-5} species.yr of damage to the ecosystem, and $227.15 worth of resource depletion. In addition, the results of uncertainty analysis of the endpoint results based on Monte Carlo simulation are attached in the last two columns of Table 3.

Fig. 5a. illustrates the results of the damage evaluation percentages, indicating that the mining and beneficiation stages marginally outperform the electrolysis stage in the resources theme. Nevertheless, the electrolysis stage dominates with an 80 % contribution in both the human health and ecosystems themes. Despite a slightly higher share in the human health and ecosystems themes, the electrolysis stage remains the overall dominant contributor. Fig. 5b. showcases the normalized results at the endpoint level, highlighting major substance contributions. It is evident that the human health theme holds dominance, with sulfur dioxide and carbon dioxide emerging as the most significant substances due to the extensive use of sulfuric acid in the electrolysis process and the energy-intensive nature of the EMM industry. Additionally, PM2.5 ranks as the third-highest impact in the human health theme. Within the ecosystems theme, sulfur dioxide and carbon dioxide maintain their dominance. Manganese stands out as the most critical substance concerning resource depletion, and the impact of coal from thermal power generation is also noteworthy.

3.6. Sensitivity analysis

Sensitivity analysis serves as a crucial reference for scientific decision-making, offering insights into the impact of parameter variations. It is an analytical tool that assesses the degree of change in results by reducing input substances by 5 %. Two rounds of sensitivity analysis were conducted. The first round explores the influence of the mining and beneficiation stages, as well as the electrolysis stage, on key impact categories collectively. The second round examines the impact of the grouping process on key impact categories. Additionally, mineral resource scarcity is considered, expanding the scope beyond the initially mentioned five impact categories. The results of the initial round of sensitivity analyses are presented in Fig. 6a. These results indicate that a 5 % reduction in overall material inputs during the electrolysis stage would lead to a nearly 4 % decrease in impacts for key categories, including terrestrial ecotoxicity, marine ecotoxicity, fossil resource scarcity, and global warming. Furthermore, a 5 % reduction in overall material inputs to the mining and beneficiation stages would particularly alleviate concerns related to mineral resource scarcity. The results of the second round of sensitivity analysis are presented in Table 4, providing specific information on the changes in various key categories as the key process substance input is reduced by 5 %. Fig. 6b. visually highlights the sensitive processes. For instance, a 5 % reduction in electricity input results in a decrease of 238.88 kg CO_2 eq in global warming and a reduction of 46.09 kg oil eq in fossil energy consumption. Additionally, reducing ammonia input by 5 % during EMM production

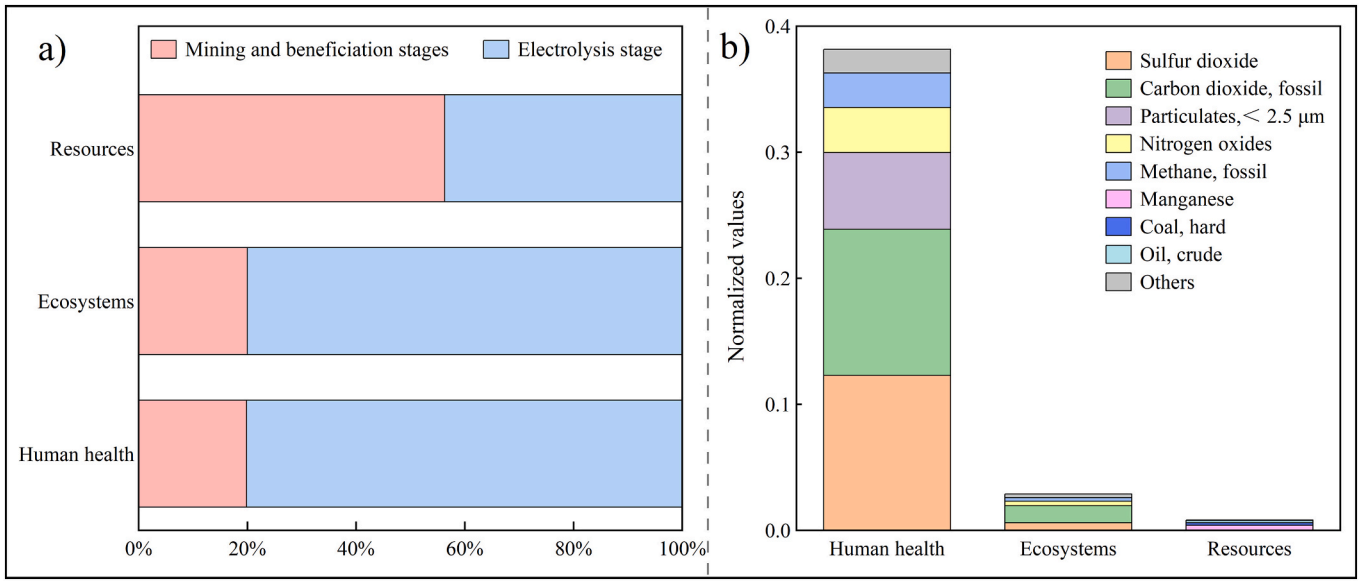


Fig. 5. LCIA endpoint results: a) proportion of damage evaluation of endpoint results; b) the key contributors in normalized endpoint results.

leads to a 1.68 % and 1.67 % decrease in the environmental burdens of terrestrial ecotoxicity and marine ecotoxicity, respectively, with no impact on mineral resource scarcity. Therefore, optimizing the power structure and improving electricity consumption efficiency emerge as effective measures for mitigating the environmental impacts of EMM production.

4. Discussion

Comparing research results with other studies helps validate the reliability of the research, provide context, and offer new insights and directions for future research. Table 5 compares the primary distinctions between the current study and previous studies. In studies conducted in China, manganese ore grades remained below 20 %, in stark contrast to the exceptionally high grades of up to 44 % observed in South African counterparts. As a result, the Chinese research exhibited elevated levels of solid waste generation and electricity consumption compared to South African (Peng et al., 2011). With minimal variance in ore grade, this study produces a higher volume of solid waste compared to other research endeavors in China, although the disparity in power consumption is relatively inconspicuous (Ning et al., 2010). The observed higher global warming impact in this study compared to the South African study is attributed to its greater electricity consumption. On the other hand, when electricity consumption is similar, this study exhibits a

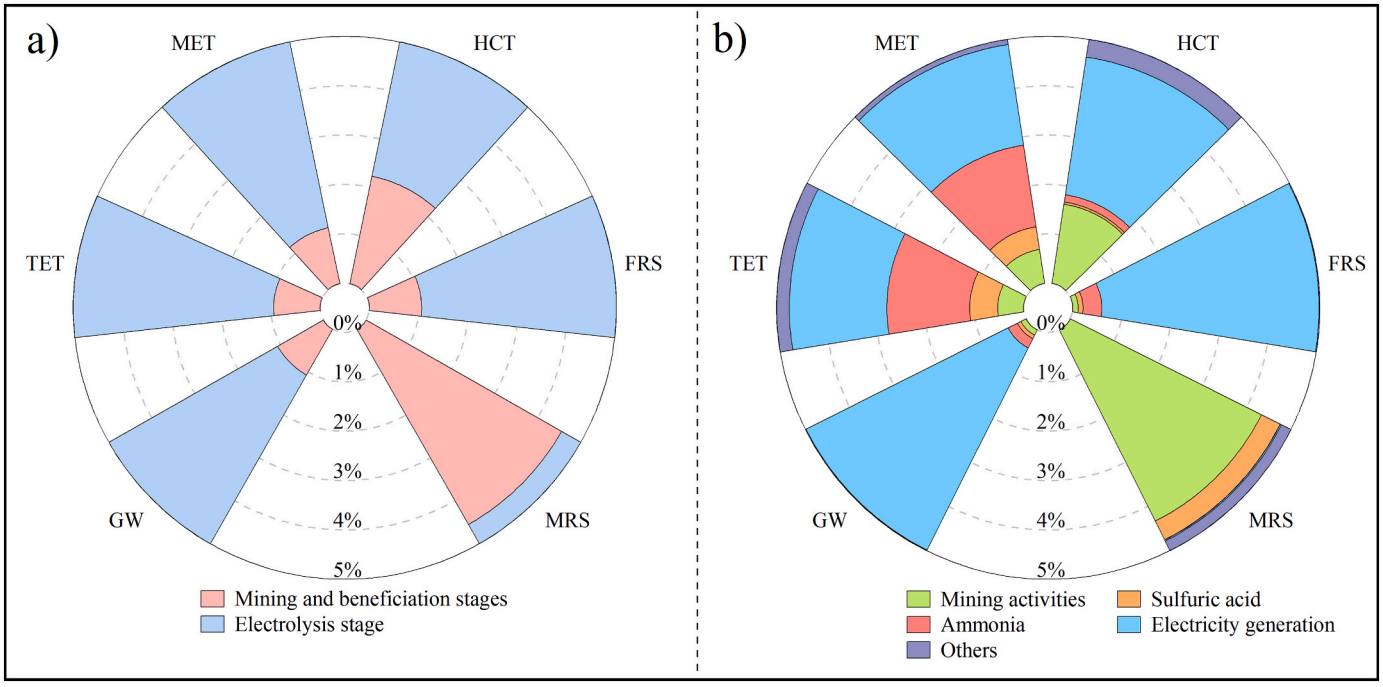


Fig. 6. Results of sensitivity analysis: a) sensitivity analysis of mining and beneficiation stages and electrolysis stage; b) Sensitivity analysis of key processes (HCT: human carcinogenic toxicity; TET: terrestrial ecotoxicity; MET: marine ecotoxicity; FRS: fossil resource scarcity; GW: global warming; MRS: mineral resource scarcity).

Table 4
Sensitivity analysis of key processes.

Category	Unit	Mining activities (RR:5 %)		Sulfuric acid (RR:5 %)		Electricity (RR:5 %)		Ammonia (RR:5 %)	
		RV	RR	RV	RR	RV	RR	RV	RR
HCT	kg 1,4-DCB eq	0.62	1.62 %	0.02	0.05 %	1.08	2.82 %	0.06	0.15 %
TET	kg 1,4-DCB eq	29.91	0.52 %	32.46	0.57 %	113.22	1.98 %	95.95	1.68 %
MET	kg 1,4-DCB eq	0.03	0.70 %	0.02	0.47 %	0.10	2.06 %	0.08	1.67 %
FRS	kg oil eq	1.13	0.11 %	1.06	0.10 %	46.09	4.39 %	3.97	0.38 %
GW	kg CO_2 eq	6.04	0.12 %	4.26	0.08 %	238.88	4.58 %	10.84	0.21 %
MRS	kg Cu eq	11.65	4.33 %	1.13	0.42 %	0.07	0.02 %	0	0

Table 5
Comparisons with previous studies.

	Unit	South Africa (Peng et al., 2011)	China a (Ning et al., 2010)	China b (Peng et al., 2011)	China c (Zhang et al., 2020)	This study
Mn ore content	%	44	16	18	19	19
Solid waste	t	1.67	7.68	6.5	4.31	9.25
H_2SO_4 consumption	t	0.25	1.96	1.52	0.81	0.98
Electricity	kWh	2100	6800	6800	7385	7675
Global warming	t CO_2 eq	2.61	–	7.85	9.59	5.21
Acidification	kg SO_2 eq	18.9	–	56.7	17.9	28.96

lower global warming impact compared to other studies conducted in China (Zhang et al., 2020). This can be explained by the detailed modeling in this research, which accounts for the specific electricity structure of Guizhou Province. In 2020, Guizhou's electricity mix comprised 61.14 % thermal power and 32.67 % hydropower (CEC, 2020). Furthermore, the sulfuric acid consumption in this study surpassed that of both South Africa and China c, resulting in a higher acidification value compared to these two studies.

According to Sections 3.2 and 3.3, it is evident that the environmental impact of the electrolysis stage surpasses that of the mining and beneficiation stages. Concurrently, the consumption of sulfuric acid and ammonia during the electrolysis stage imposes a significant burden on the environment. As mineral resources deplete and the manganese ore grade decreases, there is a growing consumption of sulfuric acid and ammonia, leading to an increase in electrolytic manganese residue during the electrolysis process (Fu et al., 2023). Owing to the extended absence of innovation and enhancement in China's manganese electrolysis process, the outdated method persists, leading to significant consumption of sulfuric acid and ammonia (Duan et al., 2011). Therefore, there is a pressing demand to overhaul the manganese electrolysis production process. Concurrently, it is imperative to devise and implement recycling systems aimed at optimizing the recovery and reuse of sulfuric acid and ammonia (Wang et al., 2023). Table 5 illustrates that the quantity of solid waste in this case surpasses that in other studies, highlighting the pressing need to address the challenges of electrolytic manganese residue and mine tailings stockpiling within this specific enterprise. Typically, the production of 1 t of EMM results in the generation of 10–12 t of electrolytic manganese residue, containing not only soluble manganese but also a significant quantity of harmful ions like Cu^{2+}, Cr^{6+}, Cd^{2+}, Se^{4+}, and others (Chen et al., 2020; D.J. He et al., 2021). In 2022, China's reserve of electrolytic manganese residue is estimated to be around 130 million tonnes, with an annual increase of approximately 10 million tonnes (Su et al., 2023; Wang et al., 2023). Without proper treatment, these residues can pose significant ecological and environmental risks to surrounding soil and surface water bodies (Yang et al., 2022). Therefore, ensuring the harmless treatment of electrolytic manganese residue is of paramount importance. Solidification stands out as a primary approach for handling toxic and hazardous solid wastes. The conventional technique for treating electrolytic manganese residue involves introducing alkaline reagents to create an alkaline environment (Li et al., 2021). This environment prompts the manganese element to undergo a reaction, resulting in the formation of an insoluble substance, ultimately establishing a stable seal (Tao et al., 2009). Effectively recycling and utilizing electrolytic manganese residue requires the crucial task of separating and extracting harmful components (Li et al., 2023). Common methods for separating harmful substances from electrolytic manganese residue include water washing, hydrothermal washing, acid washing, bioleaching, and electrochemical leaching (Wang et al., 2023). The leaching process is predominantly influenced by the valence state of metal elements and the mineral phase in the residues. Regarding the leached components, electrolytic manganese residue exhibits significantly higher leaching rates for manganese and ammonia-nitrogen compared to other residues (Shu et al., 2022). Recycling leached manganese from electrolytic manganese residue offers a sustainable solution, reducing the need for additional manganese mining. Additionally, ammonia nitrogen can be effectively contained through precipitation with ammonium magnesium phosphate (Chen et al., 2016). Another approach involves optimizing the manganese electrolysis production process to eliminate harmful substances from the overall procedure (Wang et al., 2023). The research object is located in southwest China, where infrastructure is being improved and large quantities of construction materials are needed (She et al., 2022). For sustainability, electrolytic manganese residue can be repurposed in the production of construction materials (S.C. He et al., 2021). Specifically, it can serve as a raw material for sulfur concrete. The resulting sulfur concrete samples exhibit remarkable mechanical strength and environmental durability. Importantly, the leaching concentration of heavy metal ions is significantly below China's National Comprehensive Wastewater Emission Standard (GB 8978-1996) limit values (Yang et al., 2014). Hence, a rational, effective, and site-specific treatment of electrolytic manganese residue would greatly mitigate the environmental impacts arising from the electrolysis process during production.

As demonstrated in Section 3.4, electricity generation significantly contributes to pollution in the EMM production process, a phenomenon inseparable from China's energy structure marked by a heavy reliance on coal, limited oil reserves, and a scarcity of natural gas (Xie et al., 2010). In 2022, China's electricity generation from thermal power plants reached 5.73×10^{12} kWh, accounting for 65.90 % of the nation's total power output (CEC, 2023). Primarily fueled by coal combustion, over 90 % of China's thermal power generation contributes to elevated emissions of harmful pollutants, including particulate matter, sulfur oxides, nitrogen oxides, and carbon oxides. This emission profile raises serious concerns regarding its impact on both human health and the environment (Larki et al., 2023). In response to the climate change impact of coal combustion, China has set ambitious targets, aiming to peak carbon emissions by 2030 and achieve carbon neutrality by 2060

(Gallagher et al., 2019). In 2022, China achieved a non-fossil energy electricity generation of 3.14×10^{12} kWh. Within this figure, hydropower contributed 1.35×10^{12} kWh, constituting 15.5 %, while wind power supplied 7.62×10^{11} kWh, accounting for 8.8 % (CEC, 2023). With a continued emphasis on expanding its clean energy generation capacity, China has the potential to significantly mitigate the environmental impact associated with electricity generation. If clean energy entirely replaces conventional electricity in this case study, a noticeable difference in the results emerges. Fig. 7. illustrates the environmental impact comparison at the midpoint of scenarios where wind power and hydropower replace actual electricity generation. The results demonstrate that these simulated scenarios, utilizing either wind power or hydropower, result in a reduction of over 90 % in various midpoint environmental impacts, including global warming, fine particulate matter formation, and fossil resource scarcity, compared to the actual electricity generation scenarios in the case. This underscores the significant positive impact that transitioning to clean energy sources can have on mitigating environmental concerns in electricity generation. In recent years, due to the policy support from the Chinese government, the scale of wind power generation in China has become the world's largest (Zhou et al., 2023). However, issues such as wind power curtailment and blind investments persist (Chen et al., 2023). Therefore, it is crucial to intensify technological innovation within the wind power industry. Simultaneously, a comprehensive approach should be taken, considering the development of wind power in conjunction with the specific conditions of the local area, including the characteristics and geographic location of wind power. Lastly, exploring a strategy of "compensate for each other's strengths and weaknesses" could be considered, actively promoting cooperation between regions abundant in wind resources and those lacking in such resources (Han et al., 2023). In this simulation, hydropower has shown superior sustainability compared to wind power and is considered China's most promising renewable energy source (Xie et al., 2019). Globally, hydropower constitutes 80 % of renewable energy generation capacity due to its inherent natural advantages, cost-effectiveness, and minimal pollution emissions, positioning it as a transitional bridge between traditional and alternative energy sources (Galletti et al., 2021; Trottier et al., 2022). However, 71 % of China's hydropower resources are concentrated in the southwest region (Li et al., 2018). Therefore, the development of hydropower also needs to be tailored to local conditions. In summary, replacing the power source for EMM production can significantly reduce the environmental impact associated with electricity generation.

5. Conclusions

As the economy develops, the demand for EMM is on the rise. However, the production of EMM is accompanied by severe pollution, posing continuous threats to the ecological environment and human health. In order to quantify the environmental impact of EMM production, the LCA method was employed to analyze the mining, beneficiation, and electrolysis processes of a typical EMM production enterprise in the Manganese Triangle region of China.

Midpoint results, along with uncertainties, were obtained and normalized. The normalized outcomes revealed that the primary environmental impact categories were Human carcinogenic toxicity, Terrestrial ecotoxicity, Marine ecotoxicity, Human non-carcinogenic toxicity, and Fossil resource scarcity. These impacts were primarily attributed to the electrolysis stage. Additionally, a grouped midpoint characterization analysis and identification of key factors were conducted. The results indicate that power generation and mining activities contribute 65.3 % and 24.4 % of the human carcinogenic toxicity in the production of EMM, respectively. Chromium (VI) is the main hazardous substance from both electricity generation and mining activities, contributing to 91 % of the human carcinogenic toxicity. The endpoint results indicate that producing 1 t of EMM inflicts damage on human health, totaling 1.01×10^{-2} DALY. Simultaneously, it adversely affects the ecological environment, amounting to 1.97×10^{-5} species.yr, and leads to resource depletion valued at $227.15. Additionally, sensitivity analysis was conducted, and the results showed that reducing power input by 5 % can reduce the toxicity of 1.08 kg 1,4-DCB eq in human carcinogenic toxicity. This study also affirms that substituting with clean energy can notably improve the environmental benefits of EMM production. It must be pointed out that there are some limitations. After all, the research data only comes from some regions. Although this region is rich in manganese resources, it still cannot represent the situation of other regions. In the future, more comprehensive information should be collected and research should be conducted at the national level.

For achieving cleaner production of EMM, it is recommended, based on the above analysis, to implement practical management measures guided by environmental science to address environmental issues in EMM production. The EMM production process demands innovation to reduce the reliance on feedstocks such as sulfuric acid and ammonia.

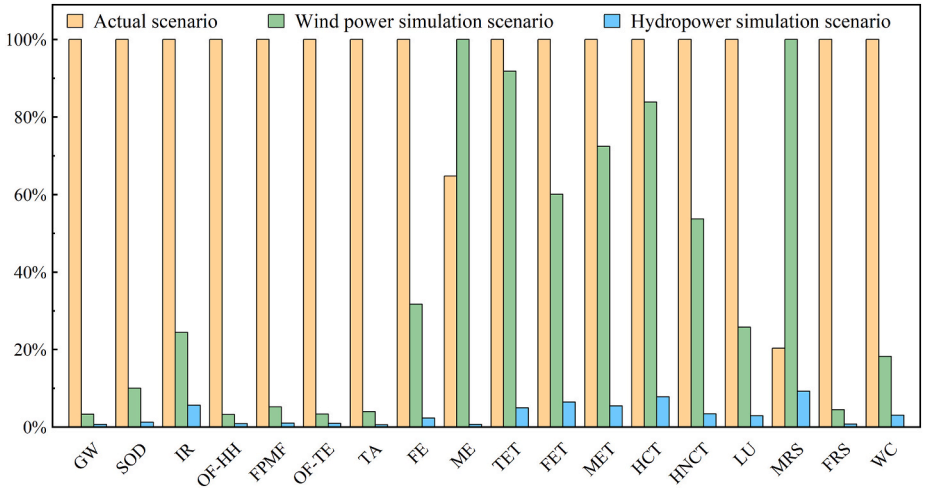


Fig. 7. The environmental impact comparison between hydropower simulation scenario, wind power simulation scenario, and the actual scenario (GW: global warming; SOD: stratospheric ozone depletion; IR: ionizing radiation; OF-HH: ozone formation-human health; FPMF: fine particulate matter formation; OF-TE: ozone formation-terrestrial ecosystems; TA: terrestrial acidification; FE: freshwater eutrophication; ME: marine eutrophication; TET: terrestrial ecotoxicity; FET: freshwater ecotoxicity; MET: marine ecotoxicity; HCT: human carcinogenic toxicity; HNCT: human non-carcinogenic toxicity; LU: land use; MRS: mineral resource scarcity; FRS: fossil resource scarcity; WC: water consumption).

The treatment of electrolytic manganese residue through alkaline sealing, leaching of harmful substances, or application in the field of building materials is an effective measure for environmentally friendly disposal. Looking ahead, China must actively promote energy transition and expedite the development of renewable energy sources. Enterprises should enhance their electricity efficiency.

CRediT authorship contribution statement

Ming Tao: Writing – review & editing, Resources, Project administration, Conceptualization. **Xianpeng Qiu:** Writing – original draft, Validation. **Daoming Lu:** Validation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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